FCFF & FCFE DCF Valuation Model

Illustrative teaching model — Reliance Industries (FCFF) and HDFC Bank (FCFE)

How this workbook is organized

Read Me	This sheet — legend, structure, and disclaimers.
Reliance FCFF	Free Cash Flow to Firm model: EBIT build → NOPAT → FCFF → WACC → Enterprise Value → Equity Value → Share Price.
HDFC Bank FCFE	Free Cash Flow to Equity model: Net Income build → FCFE → Cost of Equity (CAPM) → Equity Value directly → Share Price.
Comparison	Side-by-side summary of both models' key drivers and outputs, linked live from the two model sheets.

Color legend (standard financial-modeling convention)

Sample	Hardcoded input — the only cells you should change. Flex growth rates, WACC drivers, capex, etc. here.
Sample	Formula — calculated automatically. Do not overwrite; it will break the chain.
Sample	Link pulling a value from another sheet (used on the Comparison tab).

Important disclaimer

All financial figures in this workbook (EBIT, Net Income, D&A, capex, working-capital changes, net debt, betas, risk-free rate, etc.) are ILLUSTRATIVE, ROUND-NUMBER ESTIMATES built for teaching how FCFF/FCFE and DCF mechanics fit together — they are NOT pulled from Reliance Industries' or HDFC Bank's actual financial statements and should NOT be used for real investment decisions. To turn this into a real model, replace every yellow/blue input cell with figures sourced from the companies' latest annual reports (10-K/Annual Report equivalents), stock exchange filings, and a live cost-of-capital data provider, and cite the source next to each cell.

Reliance Industries — FCFF / WACC / Enterprise Value Model (₹ Crore, illustrative)

1. Base-Year (Year 0) FCFF Build

EBIT (Year 0)	$90,000
Tax Rate	25.0%
NOPAT = EBIT x (1 - Tax Rate)	**$67,500**
Add: Depreciation & Amortization	$35,000
Less: Capital Expenditure (Capex)	$55,000
Less: Increase in Net Working Capital	$5,000
FCFF (Year 0)	**$42,500**

2. Weighted Average Cost of Capital (WACC)

Risk-free Rate (Rf)	7.0%
Beta	1.10
Equity Risk Premium (ERP)	6.0%
Cost of Equity (Re) = Rf + Beta x ERP	**13.6%**
Pre-tax Cost of Debt (Rd)	8.0%
After-tax Cost of Debt = Rd x (1 - Tax Rate)	6.0%
Weight of Equity (E / V)	75.0%
Weight of Debt (D / V)	25.0%
Check: Weights sum to 100%	OK
WACC = We x Re + Wd x Rd(after-tax)	**11.7%**

3. 5-Year FCFF Projection

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
FCFF Growth Rate (y/y)		10.0%	9.0%	8.0%	7.0%	6.0%
FCFF	**$42,500**	**$46,750**	**$50,958**	**$55,034**	**$58,886**	**$62,420**
Discount Factor = 1 / (1+WACC)^n	1.000	0.895	0.801	0.718	0.642	0.575
PV of FCFF		$41,853	$40,842	$39,489	$37,827	$35,897

4. Terminal Value

Terminal Growth Rate (g)	5.0%
Terminal Value (at end of Year 5) = FCFF5 x (1+g) / (WACC - g)	**$978,219**
PV of Terminal Value	**$628,381**

5. Equity Value Per Share

Sum of PV of FCFF (Years 1-5)	$195,907
PV of Terminal Value	$628,381
Enterprise Value	**$824,289**
Less: Net Debt	$100,000
Equity Value	**$724,289**
Shares Outstanding (Crore)	676.00
Implied Value per Share (₹)	**$1,071.43**

(₹ Crore equity value ÷ shares outstanding in Crore = ₹ per share — units cancel directly)

HDFC Bank — FCFE / Cost of Equity / Equity Value Model (₹ Crore, illustrative)

1. Base-Year (Year 0) FCFE Build

Net Income (Year 0)	$60,000
Add: Depreciation & Amortization	$2,000
Less: Capital Expenditure (Capex)	$3,000
Less: Net Investment in Loan Book / Work	$8,000
Add: Net Borrowing (change in subordina	$1,000
FCFE (Year 0)	**$52,000**

Note: for banks, 'working capital' is replaced by loan-book growth, and regulatory (Basel/RBI) capital retention needs often further reduce distributable FCFE.

2. Cost of Equity (CAPM) — the FCFE discount rate

Risk-free Rate (Rf)	7.0%
Beta	0.90
Equity Risk Premium (ERP)	6.0%
Cost of Equity (Re) = Rf + Beta x ERP	**12.4%**

Note: no WACC here — a bank's 'debt' (deposits/borrowings) is raw material for its core business, not a separate financing layer, so there is no meaningful Enterprise Value / WACC step.

3. 5-Year FCFE Projection

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
FCFE Growth Rate (y/y)		14.0%	13.0%	12.0%	11.0%	10.0%
FCFE	**$52,000**	**$59,280**	**$66,986**	**$75,025**	**$83,277**	**$91,605**
Discount Factor = 1 / (1+Re)^n	1.000	0.890	0.792	0.704	0.627	0.557
PV of FCFE		$52,740	$53,022	$52,833	$52,175	$51,061

4. Terminal Value

Terminal Growth Rate (g)	6.0%
Terminal Value (at end of Year 5) = FCFE	**$1,517,212**
PV of Terminal Value	**$950,563**

5. Equity Value Per Price

Sum of PV of FCFE (Years 1-5)	$261,831
PV of Terminal Value	$950,563
Equity Value	**$1,212,394**
Shares Outstanding (Crore)	555.00
Implied Value per Share (₹)	**$2,184.49**

(₹ Crore equity value ÷ shares outstanding in Crore = ₹ per share — units cancel directly)

FCFF vs FCFE — Side-by-Side Comparison

	Reliance Industries (FCFF)	HDFC Bank (FCFE)
Valuation method	FCFF discounted at W	FCFE discounted at Cost of Equity
Discount rate used	11.7%	12.4%
Base-year cash flow (₹ Cr)	$42,500	$52,000
Terminal growth rate	5.0%	6.0%
Sum PV of explicit cash flows (₹ Cr	$195,907	$261,831
PV of Terminal Value (₹ Cr)	$628,381	$950,563
Enterprise Value (₹ Cr)	$824,289	n/a — banks are valued directly to equity
Less: Net Debt (₹ Cr)	$100,000	n/a
Equity Value (₹ Cr)	$724,288.62	$1,212,394.09
Implied Value per Share (₹)	$1,071	$2,184

Why the methods differ

- Reliance: leverage has swung a lot (Jio debt buildout, then large equity stake sales), so FCFF/WACC values the whole enterprise first, and net debt is
- HDFC Bank: deposits/borrowings are the core business, not just funding, so there is no clean 'pre-interest' operating line — FCFE is discounted straight to
- If assumptions were fully consistent, an FCFF-based equity value and an FCFE-based equity value for the SAME company would converge — divergence